

## Introduction to Embedded Systems

### What is Embedded System?

An Electronic/Electro mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware (Software)

e.g: Electronic Toys, Mobile Handsets, Washing Machines, Air Conditioners, Automotive Control Units, Set Top Box, DVD Player etc...

**Embedded Systems are:**

- ☐ Unique in character and behavior
- ☐ With specialized hardware and software

### Embedded Systems Vs General Computing Systems:

| Criteria          | General Purpose Computing System                                                                                                      | Embedded System                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Contents          | A system which is a combination of a generic hardware and a General Purpose Operating System for executing a variety of applications. | A system which is a combination of special purpose hardware and embedded OS/firmware for executing a specific set of applications |
| OS                | It contains a general purpose operating system (GPOS).                                                                                | It may or not contain an operating system for functioning.                                                                        |
| Alterations       | <u>Applications</u> are alterable by the user.                                                                                        | <u>Applications</u> are not-alterable by the user.                                                                                |
| Key factor        | Performance is key factor.                                                                                                            | Application specific requirements are key factors.                                                                                |
| Power Consumption | More                                                                                                                                  | Less                                                                                                                              |
| Response Time     | Not critical                                                                                                                          | Critical for some applications                                                                                                    |
| Execution Time    | Need not be deterministic                                                                                                             | Deterministic for certain types of ES                                                                                             |

## Classification of Embedded Systems:

- ☐ Based on Generation
- ☐ Based on Complexity & Performance Requirements
- ☐ Based on deterministic behavior
- ☐ Based on Triggering

### 1. Embedded Systems - Classification based on Generation

- **First Generation:** The early embedded systems built around 8-bit microprocessors like 8085 and Z80 and 4-bit microcontrollers  
**Eg: stepper motor control units, Digital Telephone Keypads etc.**
- **Second Generation:** Embedded Systems built around 16-bit microprocessors and 8 or 16-bit microcontrollers, following the first generation embedded systems  
**Eg: SCADA, Data Acquisition Systems etc.**
- **Third Generation:** Embedded Systems built around high performance 16/32 bit Microprocessors/controllers, Application Specific Instruction set processors like Digital Signal Processors (DSPs), and Application Specific Integrated Circuits (ASICs).The instruction set is complex and powerful.  
**Eg: Robotics, industrial process control, networking etc.**
- **Fourth Generation:** Embedded Systems built around System on Chips (SoCs), Re-configurable processors and multi core processors. It brings high performance, tight integration and miniaturization into the embedded device market  
**Eg: Smart phone devices, MIDs etc.**

### 2. Embedded Systems - Classification based on Complexity & Performance

- **Small Scale:** The embedded systems built around low performance and low cost 8 or 16 bit microprocessors/ microcontrollers. It is suitable for simple applications and where performance is not time critical. It may or may not contain OS.
- **Medium Scale:** Embedded Systems built around medium performance, low cost 16 or 32 bit microprocessors / microcontrollers or DSPs. These are slightly complex in hardware and firmware. It may contain GPOS/RTOS.
- **Large Scale/Complex:** Embedded Systems built around high performance 32 or 64 bit RISC processors/controllers, multi-core processors and Programmable Logic Devices (PLD). It requires complex hardware and software. These system may contain multiple processors/controllers and co-units/hardware accelerators for offloading the processing requirements from the main processor. It contains Real Time OS for scheduling, prioritization and management.

### 3. Embedded Systems - Classification Based on deterministic behavior:

It is applicable for Real Time systems. The application/task execution behavior for an embedded system can be either deterministic or non-deterministic

**These are classified in to two types**

1. **Soft Real time Systems:** Missing a deadline may not be critical and can be tolerated to a certain degree
2. **Hard Real time systems:** Missing a program/task execution time deadline can have catastrophic consequences (financial, human loss of life, etc.)

### 4. Embedded Systems Classification Based on Triggering:

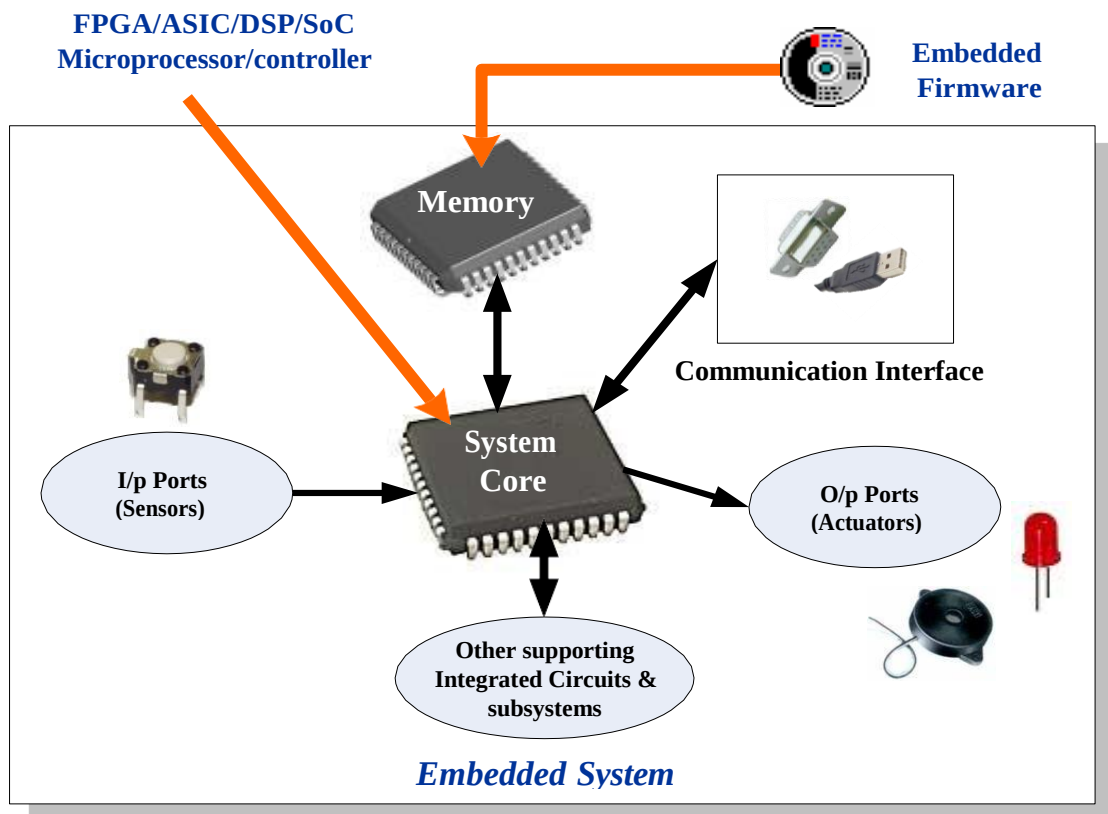
**These are classified into two types**

1. **Event Triggered:** Activities within the system (e.g., task run-times) are dynamic and depend upon occurrence of different events.
2. **Time triggered:** Activities within the system follow a statically computed schedule (i.e., they are allocated time slots during which they can take place) and thus by nature are predictable.

### Elements of Embedded Systems:

An embedded system is a combination of 3 things, Hardware Software Mechanical Components and it is supposed to do one specific task only. A typical embedded system contains a single chip controller which acts as the master brain of the system. Diagrammatically an embedded system can be represented as follows:

Embedded systems are basically designed to regulate a physical variable (such Microwave Oven) or to manipulate the state of some devices by sending some signals to the actuators or devices connected to the output port system (such as temperature in Air Conditioner), in response to the input signal provided by the end users or sensors which are connected to the input ports. Hence the embedded systems can be viewed as a reactive system. The control is achieved by processing the information coming from the sensors and user interfaces and controlling some actuators that regulate the physical variable.



### ***Real World***

Keyboards, push button, switches, etc. are Examples of common user interface input devices and LEDs, LCDs, Piezoelectric buzzers, etc examples for common user interface output devices for a typical embedded system. The requirement of type of user interface changes from application to application based on domain.

Some embedded systems do not require any manual intervention for their operation. They automatically sense the input parameters from real world through sensors which are connected at input port. The sensor information is passed to the processor after signal conditioning and digitization. The core of the system performs some predefined operations on input data with the help of embedded firmware in the system and sends some actuating signals to the actuator connect connected to the output port of the system.

The memory of the system is responsible for holding the code (control algorithm and other important configuration details). There are two types of memories are used in any embedded system. Fixed memory (ROM) is used for storing code or program. The user cannot change the firmware in this type of memory. The most common types of memories used in embedded systems for control algorithm storage are OTP,

PROM, UVEPROM, EEPROM and FLASH

An embedded system without code (i.e. the control algorithm) implemented memory has all the peripherals but is not capable of making decisions depending on the situational as well as real world changes. Memory for implementing the code may be present on the processor or may be implemented as a separate chip interfacing the processor. In a controller based embedded system, the controller may contain internal memory for storing code such controllers are called Micro-controllers with on-chip ROM, eg. Atmel AT89C51.

### The Core of the Embedded Systems:

#### General Purpose and Domain Specific Processor:

- Almost 80% of the embedded systems are processor/ controller based.
- The processor may be microprocessor or a microcontroller or digital signal processor, depending on the domain and application.

#### **Microprocessor:**

- A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions, which is specific to the manufacturer
- In general the CPU contains the Arithmetic and Logic Unit (ALU), Control Unit and Working registers
- Microprocessor is a dependant unit and it requires the combination of other hardware like Memory, Timer Unit, and Interrupt Controller etc for proper functioning.
- Intel claims the credit for developing the first Microprocessor unit Intel 4004, a 4 bit processor which was released in Nov 1971.

#### **Microcontroller:**

- ❖ A highly integrated silicon chip containing a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports
- ❖ Microcontrollers can be considered as a super set of Microprocessors
- ❖ Microcontroller can be general purpose (like Intel 8051, designed for generic applications and domains) or application specific (Like Automotive AVR from Atmel Corporation. Designed specifically for automotive applications)
- ❖ Since a microcontroller contains all the necessary functional blocks for independent working, they found greater place in the embedded domain in place of microprocessors
- ❖ Microcontrollers are cheap, cost effective and are readily available in the market

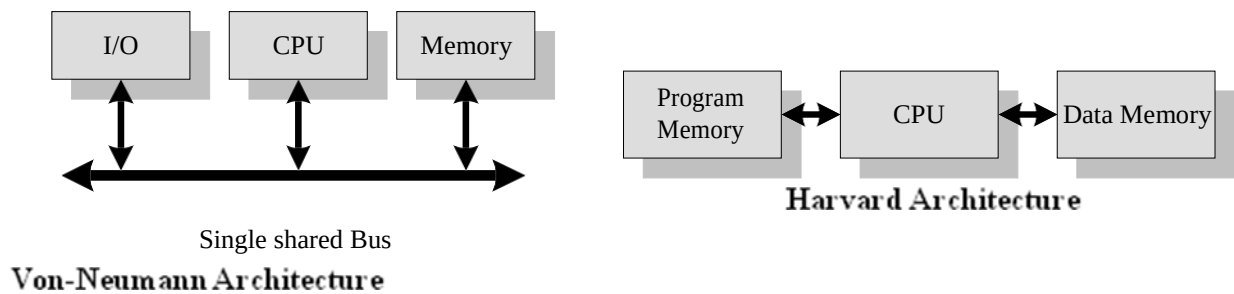
- ❖ Texas Instruments TMS 1000 is considered as the world's first microcontroller

### Microprocessor Vs Microcontroller:

| Microprocessor                                                                                                                                                                      | Microcontroller                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions | A microcontroller is a highly integrated chip that contains a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports |
| It is a dependent unit. It requires the combination of other chips like Timers, Program and data memory chips, Interrupt controllers etc for functioning                            | It is a self contained unit and it doesn't require external Interrupt Controller, Timer, UART etc for its functioning                                                                                                                    |
| Most of the time general purpose in design and operation                                                                                                                            | Mostly application oriented or domain specific                                                                                                                                                                                           |
| Doesn't contain a built in I/O port. The I/O Port functionality needs to be implemented with the help of external Programmable Peripheral Interface Chips like 8255                 | Most of the processors contain multiple built-in I/O ports which can be operated as a single 8 or 16 or 32 bit Port or as individual port pins                                                                                           |
| Targeted for high end market where performance is important                                                                                                                         | Targeted for embedded market where performance is not so critical (At present this demarcation is invalid)                                                                                                                               |
| Limited power saving options compared to microcontrollers                                                                                                                           | Includes lot of power saving features                                                                                                                                                                                                    |

### Harvard V/s Von-Neumann Processor/Controller Architecture

- The terms Harvard and Von-Neumann refers to the processor architecture design.
- Microprocessors/controllers based on the **Von-Neumann** architecture shares a single common bus for fetching both instructions and data. Program instructions and data are stored in a common main memory
- Microprocessors/controllers based on the **Harvard** architecture will have separate data bus and instruction bus. This allows the data transfer and program fetching to occur simultaneously on both buses
- With Harvard architecture, the data memory can be read and written while the program memory is being accessed. These separated data memory and code memory buses allow one instruction to execute while the next instruction is fetched ("Pre-fetching")



### Harvard V/s Von-Neumann Processor/Controller Architecture

| Harvard Architecture                                                                                                                      | Von-Neumann Architecture                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Separate buses for Instruction and Data fetching                                                                                          | Single shared bus for Instruction and Data fetching                                                                          |
| Easier to Pipeline, so high performance can be achieved                                                                                   | Low performance Compared to Harvard Architecture                                                                             |
| Comparatively high cost                                                                                                                   | Cheaper                                                                                                                      |
| No memory alignment problems                                                                                                              | Allows self modifying codes <sup>†</sup>                                                                                     |
| Since data memory and program memory are stored physically in different locations, no chances for accidental corruption of program memory | Since data memory and program memory are stored physically in same chip, chances for accidental corruption of program memory |

### RISC V/s CISC Processors/Controllers

| RISC                                                                                                        | CISC                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lesser no. of instructions                                                                                  | Greater no. of Instructions                                                                                                                          |
| Instruction Pipelining and increased execution speed                                                        | Generally no instruction pipelining feature                                                                                                          |
| Orthogonal Instruction Set (Allows each instruction to operate on any register and use any addressing mode) | Non Orthogonal Instruction Set (All instructions are not allowed to operate on any register and use any addressing mode. It is instruction specific) |
| Operations are performed on registers only, the only memory operations are load and store                   | Operations are performed on registers or memory depending on the instruction                                                                         |
| Large number of registers are available                                                                     | Limited no. of general purpose registers                                                                                                             |

|                                                                                               |                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programmer needs to write more code to execute a task since the instructions are simpler ones | . A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC |
| Single, Fixed length Instructions                                                             | Variable length Instructions                                                                                                                                       |
| Less Silicon usage and pin count                                                              | More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.                                                  |
| With Harvard Architecture                                                                     | Can be Harvard or Von-Neumann Architecture                                                                                                                         |

### Sensors & Actuators:

- Embedded system is in constant interaction with the real world
- Controlling/monitoring functions executed by the embedded system is achieved in accordance with the changes happening to the Real World.
- The changes in the system environment or variables are detected by the sensors connected to the input port of the embedded system.
- If the embedded system is designed for any controlling purpose, the system will produce some changes in controlling variable to bring the controlled variable to the desired value.
- It is achieved through an actuator connected to the out port of the embedded system.

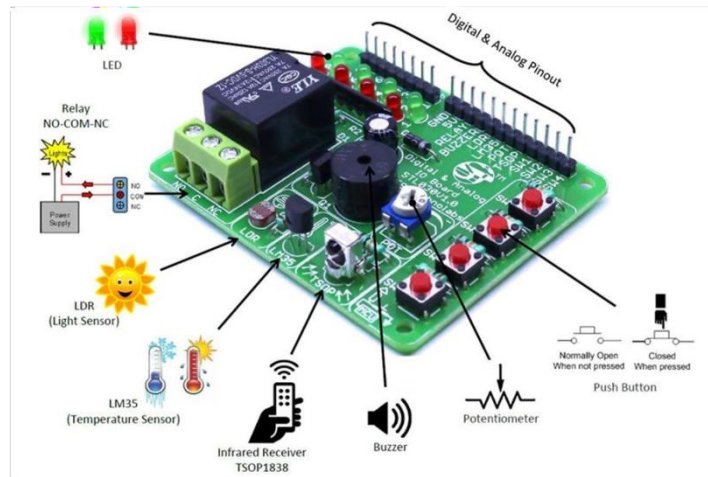
### **Sensor:**

- A transducer device which converts energy from one form to another for any measurement or control purpose. Sensors acts as input device
- Example: IR, humidity, PIR(passive infra red) , ultrasonic , piezoelectric , smoke sensors



## Actuator:

- A form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device
- Eg. Electric motor, sliding doors, Escalators, Adjusting the Car Seat



Silicon TechnoLabs Digital Analog Arduino Starter kit

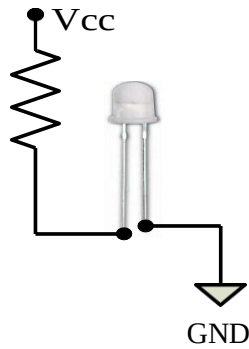
(Figure: Example to show sensors)

## The I/O Subsystem:

- The I/O subsystem of the embedded system facilitates the interaction of the embedded system with external world
- The interaction happens through the sensors and actuators connected to the Input and output ports respectively of the embedded system
- The sensors may not be directly interfaced to the Input ports, instead they may be interfaced through signal conditioning and translating systems like ADC, Opto couplers etc

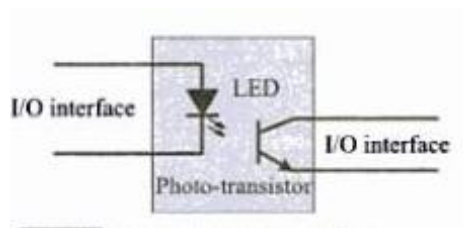
### 1. I/O Devices - Light Emitting Diode (LED):

- Light Emitting Diode (LED) is an output device for visual indication in any embedded system
- LED can be used as an indicator for the status of various signals or situations.
- For proper functioning of the LED, the anode of it should be connected to +ve terminal of the supply voltage and cathode to the -ve terminal of supply voltage
- The current flowing through the LED must limited to a value below the maximum current that it can conduct.
- A resistor is used in series between the power supply and the resistor to limit the current through the LED

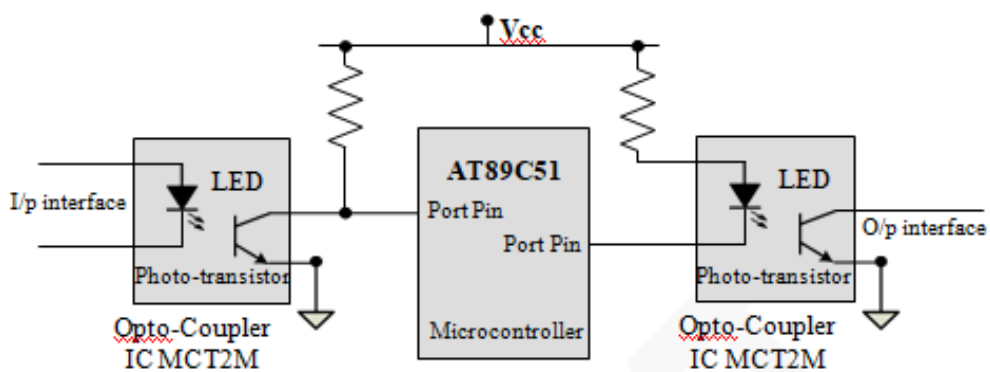


## 2. I/O Devices – **Optocoupler**

- Optocoupler is a solid state device to isolate two parts of a circuit.
- Optocoupler combines an LED and a photo-transistor in a single housing (package)
- Optocouplers can be used in either input circuits or in output circuits



- In electronic circuits, optocoupler is used for suppressing interference in data communication, circuit isolation, High voltage separation, simultaneous separation and Intensification signal etc. Below figure shows the usage of the optocoupler.



➤

## Basic Electronics - Unit 3

### Fundamentals of Communication systems

Communication is the basic process of exchange of information or transmission of information from one point to another. Basically, the information is generated from the thought process in the human mind and it is sent to the receiver in the form of speech, sign or in any other understandable form.

Communication enters our daily life in many different ways – such as telephone, radio, TV, computers....Communication provides directions for ships, aircrafts, rockets and satellites in space. A number of modern communication systems include mobile communication, point to point communication, radio telemetry and so on. Communication need not directly involve human beings always. For example communication between two or more computers, the human decision/intervention is required only while giving commands or to monitor the results.

#### Basic Blocks of Communication Systems:

Irrespective of the applications, basic blocks of electronic communication system is as shown in figure 1. This involves transmitter, channel and receiver. The transmitter is located in one point and the receiver is located somewhere else and the channel is the medium that connects them.

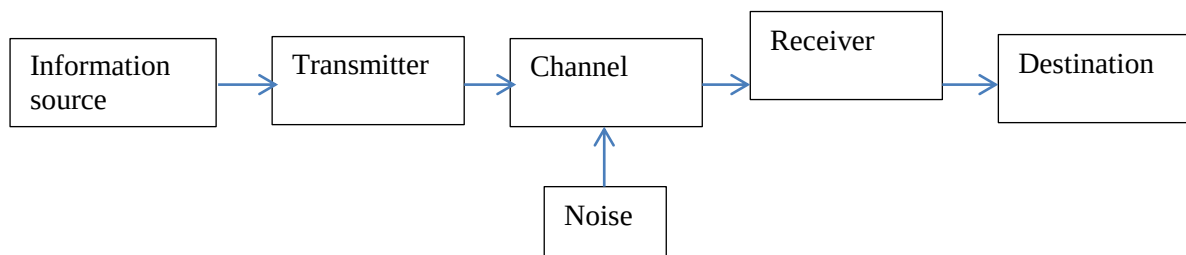


Figure 1: Block diagram of communication system

The elements of communication system are as follows:

- Information
- Transmitter
- Communication channel or medium
- Noise
- Receiver

**Information:** Information or message to be communicated are generated from information sources. Examples for main sources of information are human brain or changes in the physical environment. Information may be the speech signal, image, data music, video etc. The amount of information is measured in bits.

**Transmitter:** The information generated from a source is not in a suitable form to be transmitted directly through the channel. The transmitter is an electronic circuit designed to convert the information into a signal suitable to be transmitted over a given communication channel. Signal in its original form will be non-electrical in nature, such signal cannot be transmitted through the channel. Using suitable transducers the messages are converted to the electrical form and then they are processed and coded before the transmission. The signal processing techniques are also applied at this stage to make the communication more effective.

Transmitter includes – transducer, modulator, filters, encoder and also amplifiers to ensure faithful signal transmission and reception.

### **Communication channel:**

The communication channel is the medium through which the electronic signal is transmitted. The channel connects the transmitter and receiver either through wire or without wire. In general the communication channel can be classified as:

- Wired channel or line communication
- Wireless channel or radio channel

**Wired channel:** The examples for a wired medium are: copper wire, coaxial cable, fibre optic cable. For applications like telephony two physical wires or conductors are connected between the transmitter and the receiver. Latest applications use optical fibre cables, which are well known for their high capacity and immunity to noise. In optical fibres the information will be transmitted in the form of light wave. Coaxial cables are preferred over pair of wires as they have higher bandwidth and lower losses. Optical fibres are logical extension of coaxial cables which can be operated at higher frequencies and greater bandwidths. Optical fibres are also cheaper and are more immune to interference and noise.

**Wireless channel:** most commonly known as radio channel or electromagnetic medium. This type of channel connects the transmitter and receiver wirelessly. No physical wire is

necessary to carry the information, signal is sent through air or free space. Radio communication requires two antennas for the transmission and reception. The signal received at the receiving antenna will be attenuated and hence its amplitude will be smaller. With proper amplification, the signal will be processed further to get back the original information. Radio communication allows the signal to be transmitted to any longer distance – thousands of kilometres, perhaps even more.

**Noise:** Noise is basically any unwanted signal that disturbs the communication. Noise enters the communication system normally in the medium or channel. Noise may also be generated at the transmitter or receiver. Means noise can be internally generated in a system or externally added to the signal.

Noise can also be classified as natural noise or man-made noise. Natural noise includes – lightening during rainy season, radiations from the sun and cosmic radiations. Man-made noise is the noise generated by the electric ignition systems, industrial noise, fluorescent lights etc.

Noise imposes serious problem on electronic communication systems, if not controlled. Noise cannot be completely eliminated, but its effect can be reduced using different methods.

**Receiver:** The receiver is a collection of electronic circuits designed to convert the signal back to its original form. The process includes amplification, mixing, demodulation and decoding etc. The receiver performs the task of operating on the received signal to generate the estimation of the original signal. If the estimated signal is same as the original transmitted signal, then we say that the reception is proper.

### **Modulation:**

The transmission of an information- bearing signal over a communication channel requires a shift in the range of frequencies from the original to another frequency range suitable for transmission. This is accomplished through the process of modulation. Modulation is defined as the process by which some characteristic of a **carrier signal** is varied in accordance with a **modulating signal**. The message signal is referred to as modulating signal and the result of modulation is referred to as modulated signal. The carrier signal is usually a high frequency signal than the message signal. The message signal modifies the amplitude, frequency or phase of the carrier signal in the process of modulation. Hence, the types of modulation are:

- Amplitude modulation – amplitude of the carrier is varied in accordance with the modulating signal, keeping the frequency and phase of the carrier constant
- Frequency modulation – the frequency of the carrier signal is varied according to the modulating signal
- Phase modulation – phase of the carrier is varied in accordance with the modulating signal

### **Need for modulation:**

The message signal cannot be transmitted in its direct form, over the communication channel. It should be modulated suitably. The advantages of the modulation are:

- Reduces the antenna height
- The range(distance) of communication can be increased
- Signals can be multiplexed
- Bandwidth can be used efficiently
- Quality of reception can be improved
- Avoids mixing of the signals

**Height of the antenna:** Minimum height of the antenna for proper transmission and reception is  $\lambda/4$ , where  $\lambda$  is the wavelength. We know that,  $\lambda = c/f$ , where  $c$  is velocity of light and  $f$  is the frequency. At lower frequencies wavelength is high and antenna height is also high. Hence by modulating the message signal using a high frequency carrier, the height of the antenna can be decreased.

**Range of communication:** Since message signal frequency range is low (20 Hz to 20 kHz), the possibility of signal attenuation is more. Modulation increases the frequency and hence the range of communication may be increased.

**Multiplexing:** modulation allows the multiplexing of signals. Multiplexing means two or more message signals are transmitted simultaneously over the same channel. Eg: broadband data services, TV channels operating simultaneously, radio stations in MW and SW band simultaneously.

Bandwidth utilization: Bandwidth of the modulated signal can be made smaller or larger than the original signal.

Quality of reception: by using specific types of modulation schemes such as FM, noise performance of the receiver can be improved.

Avoids mixing: The message signal frequencies are located within the range from 20 Hz to 20 kHz. Simultaneous transmission of multiple message signals may lead to mixing of the signals, which may cause interference in the reception. Instead, if each message signal is modulated separately using different carrier frequencies, then each signal will be located at different portions in the spectrum. The receiver tuned to particular carrier will be able to receive the signal. Therefore modulation avoids mixing of signals.

## Cellular Communication

When human beings wanted to communicate between the two ends, namely sending and receiving ends, whether the distance being short or long, they needed communication devices at both the ends. Such a communication system comprises of the transmitter and receiver at the two ends. The basic block diagram of a communication system is shown in Figure 1.

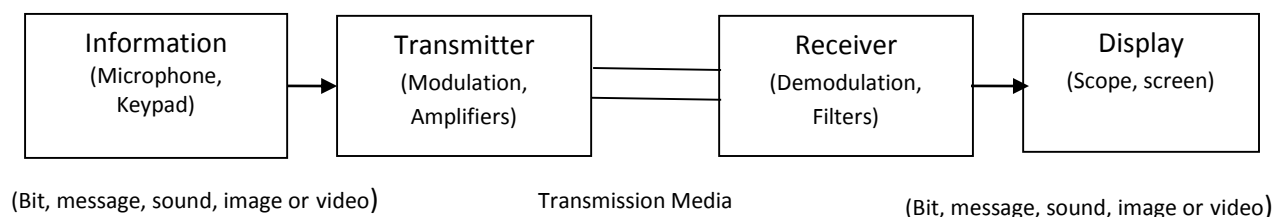


Figure 1. Communication system

The information from a microphone or keypad in the form bits, message, sound, image or video will be sent to the transmitter. The information is made ready for transmission through transmission media using the processes such as modulation, amplification etc. The transmission media may use pair of conductors, coaxial cable or optical fibres that helps in the transmission or reception of the information between the two ends. Once the information reaches the other end, it needs to be presented in the user understandable form by processes such as demodulation, filtering etc. Now, the devices such as scope or screen display the original information. Thus, the communication system helps in transmitting information from one end to the other end.

### Telecommunication:

There are many applications of communication systems, but telecommunication is the most widely used application as it connects the people around the globe. The communication device used for this communication device is 'telephone'

If the wireless communication is taking place with the movement of the communication devices, it is referred to as 'mobile communication'.

Today the telephone is replaced by a mobile phone and communication is still possible with the movement of the sending or receiving end devices or both.

### Why mobile communication is called a 'Cellular communication'?

If the mobile communication involves mobile devices and a coordinator station called base transceiver station or base station within a small geographical area, the communication is cellular communication.

Each cellular base station is allocated a group of radio channels to be used within a small geographic area called a cell.

The scheme started initially for voice transmission. But, today it has been providing internet service, transmission of audio, image and video.

**With a neat diagram, explain a cellular communication system.**

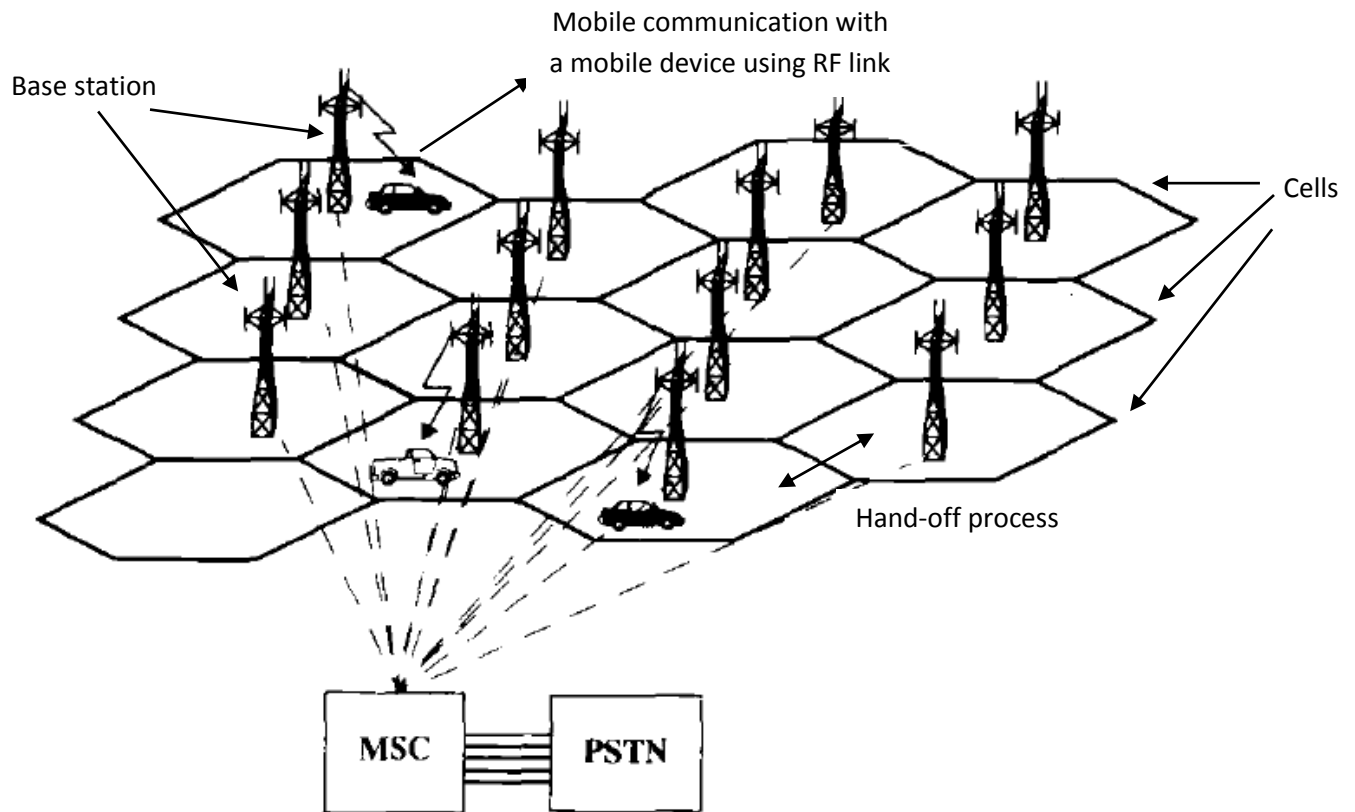


Figure 2. Cellular communication system

Figure 2 shows the block diagram of a cellular communication system in which the hexagon shaped cells are shown. The automobile (mobile user with the mobile device) shown in the figure wirelessly communicate when it moves from one cell to another cell. The base stations of the respective cell connect the mobile devices using radio Frequency (RF) link. The base stations are controlled by Mobile switching centres (MSC) which are connected to the existing telephone system referred to as Public Switched Telephone Network (PSTN). Base station is provided with a call/data communication handling processor.

**RF linkage using Base station and Mobile station:**

The entire radio frequency spectrum is divided into small frequency bands. The RF connectivity is established by the MSC with the help of Base station when a call is to be established. It assigns an available frequency band (channel), out of the entire bandwidth to the mobile unit temporarily till the end of the call.

**Hand-off process:**

Whenever a mobile user crosses one cell boundary and moves to another cell with an active call, the call is softly handed over to a new frequency band. the MSC adjusts the transmitted power of the mobile phone and changes the channel of the mobile unit and base stations in order to maintain call quality. This is called a hand-off in mobile communication.

**Explain the different events between the mobile unit and a cellular system while a phone number is entered.**

Before the MSC connects to an available frequency band, the initial handshaking is achieved by 4 different frequency bands or channels. They are shown in Table 1.

Table 1: Channels for initiating the call and voice transmission

| Type of Channels | Name of the channel           | Channels used for:                                             |
|------------------|-------------------------------|----------------------------------------------------------------|
| Control channels | Forward Control Channel (FCC) | Initiating mobile calls from the base station to mobiles.      |
|                  | Reverse Control Channel (RCC) | Initiating mobile calls from the mobiles to base station.      |
| Voice channels   | Forward Voice Channel (FVC)   | voice transmission from the base station to mobiles are called |
|                  | Reverse Voice Channel (RVC)   | voice transmission from mobiles to the base station            |

When a mobile originates a call,

1. a call initiation request is sent on the reverse control channel.
2. With this request the mobile unit transmits its telephone number- Mobile Identification Number (MIN), Electronic Serial Number (ESN), and the telephone number of the called party.
3. The mobile also transmits a station class mark (SCM) which indicates what the maximum transmitter power level is for the particular user.
4. The cell base station receives this data and sends it to the MSC.
5. The MSC validates the request, makes connection to the called party through the PSTN,
6. Now, MSC instructs the base station and mobile user to move to an unused forward and reverse voice channel pair to begin the for conversation.

**Explain the concept of ‘frequency reuse’ in a cellular system.**

Cellular systems have grown Figure 3 illustrates the concept of **cellular frequency reuse**, with the hexagon shaped cells.

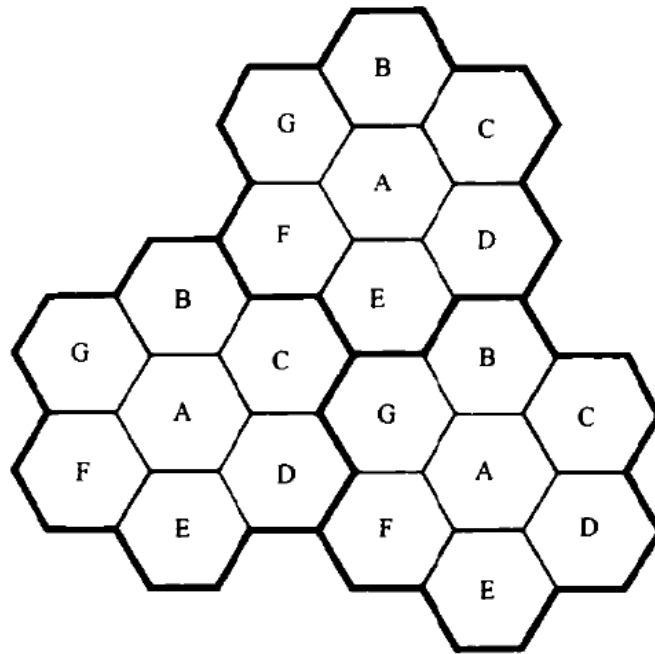


Figure 3. Cellular Frequency Reuse concept

In the Figure 3, cells numbered from A – G forms a group of cells called Cluster. There are 3 such clusters shown in the Figure, with the frequency bands repeated. Cells with the same letter use the same set of frequencies. A cell cluster is outlined in bold and replicated over the coverage area. The N cells which collectively use the complete set of available frequencies is called a cluster. From the Figure, it is clear that

1. A cluster uses frequency bands A – G is not repeated in the adjacent cells.
2. The frequency bands are repeated only in the next cluster.

The number of cells in a cluster is cluster size, denoted as ‘N’.

Frequency reuse factor is the reciprocal of cluster size.

Frequency reuse factor =  $1/N$

In this example, the **cluster size, N, is equal to seven**, and the **frequency reuse factor is 1/7** since each cell contains one-seventh of the total number of available channels.

**Why a cell is hexagon in shape?**

A cell must be designed to serve the weakest mobiles located at the edge of the cell. The hexagonal cell shape is conceptual.

Thus, when considering geometric shapes, the entire region is to be covered without overlap and with equal area,

A circle may be thought of to represent the coverage area of a base station, but adjacent circles cannot be overlaid upon a map without leaving gaps creating overlapping regions.

There are three sensible choices:

1. a square;
2. an equilateral triangle;
3. a hexagon.

Theoretically, the radiation zone cannot be represented by a square or triangle.

The hexagon permits easy and manageable analysis of a cellular system.

For a given distance between the center of a polygon and its farthest perimeter points, the hexagon has the largest area. The hexagon closely approximates a circular radiation pattern.

Using this, the fewest number of cells can cover a geographic region, which would occur for an omni-directional base station antenna.

**Note:** (i) Normally, omni-directional antennas are used in center-excited cells sectorized directional antennas are used in corner-excited cells.

(ii) The cellular frequency bands are the channels with 2 frequency bands. That's the reason they referred to as duplex channels.

(iii) If the duplex channels are based on the frequency, they are called Frequency Division Duplexing.

(iv) If the duplex channels are based on the time, they are called Time Division Duplexing.

**Explain the capacity of a cell with the necessary equations.**

**Capacity of Cluster:**

Consider a cellular system which has a total of  $S$  duplex channels (Figure 3). If each cell is allocated group of  $k$  channels ( $k < S$ ), and if the  $S$  channels are divided among  $N$  cells into unique channel groups, the total number of available radio channels will be

$$S = kN \quad (1)$$

Cluster:

If a cluster is replicated  $M$  times within the system, the total number of duplex channels,  $C$ , the capacity is given by

$$\begin{aligned} C &= MkN \\ C &= MS \quad (2) \text{ \{ Using (1) \}} \end{aligned}$$

The capacity of a cellular system is directly proportional to the number of times a cluster is replicated in a fixed service area.

The factor  $N$  is called the cluster size and is typically equal to 4, 7, or 12.

**Note:** (i) If the cluster size  $N$  is reduced while the cell size is kept constant, more clusters are required to cover a given area and hence more capacity (a larger value of  $C$ ) is achieved.

(ii) The value for  $N$  is a function of how much interference a mobile or base station can tolerate while maintaining a sufficient quality of communications.

(iii) From a design viewpoint, the smallest possible value of  $N$  is desirable in order to maximize capacity over a given coverage area (i.e.. to maximize  $C$  in equation (2)).

**Effect of Interference:**

Interference on voice channels causes cross talk,

A user hears interference in the background due to an undesired transmission is called cross talk.

On control channels, interference leads to missed and blocked calls due to errors in the digital signaling.

Interference is more severe in urban areas, due to the greater HF noise floor and the large number of base stations and mobiles.

Interference does not allow increase of capacity and there will be call drop.

Types:

- A. Co-channel interference
- B. Adjacent channel interference.

**Co-channel Interference and System Capacity**

Co-channel cells are the cells that use frequency reuse where the cells in a given geographical area use the same set of frequency. The interference between signals from these cells is called co-channel interference.

Thermal noise can be reduced by increasing the signal-to noise ratio (SNR), but co-channel interference cannot be reduced by increasing the carrier power of a transmitter. This is because an increase in carrier transmit power increases the interference to neighboring co-channel cells.

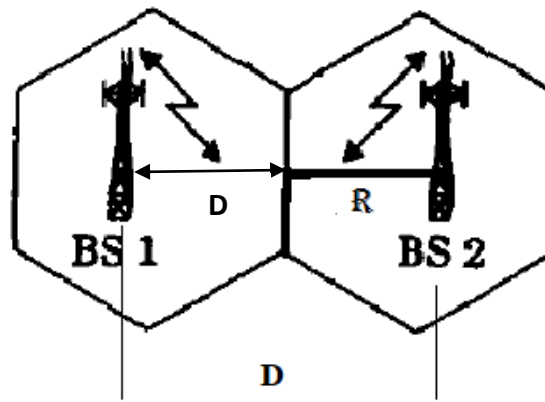


Figure 4. Adjacent co-channel cells with the parameters

Thus interference is reduced from improved isolation of high frequency energy from the co-channel cell.

Co-channel reuse ratio: (Q)

It related to the

Cluster size - N

Distance between the centers of the nearest co-channel cells - D

Radius of the cell – R

For a hexagonal geometry,

$$Q = \frac{D}{R} = \sqrt{3N} \text{ ----- (3)}$$

Smaller the Q, larger will be the capacity since N (Cluster size) is small.

Larger the Q, co-channel interference will be smaller. This improves the transmission quality of the signal.

### Problems:

1. If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the total available channels in the system.

#### Solution :

Given:

Total bandwidth = 33 MHz = 33,000 kHz.

Channel bandwidth = 25 KHz x 2 simplex channels = 50 kHz/duplex channel

Total available channels = 33,000/50 = 660 channels.

2. If there are 4 groups of channels in a cellular system with a cluster size of 7, then find the total number of duplex channels. If there are 5 clusters covering the geographical area, then find the cellular capacity.

Solution:

$k = 4$

$N = 7$

$S = ?$

We know that  $S = k \cdot N$

Therefore,  $S = 4 \times 7$

$= 28$  duplex channels.

The cellular capacity is  $C = M \cdot S = 5 \times 28 = 140$  duplex channels.

3. If there are total of 32 duplex channels in each cell in a cellular system, having a capacity of 64 duplex channels, find the number of clusters.

$M = ? \quad S = 32 \quad C = 64, \quad C = M S$

Hence, the number of clusters is  $M = C/S = 64/32 = 2$

4. If the radius of a cell is 10 km and the distance between the centers of adjacent co-channel cells is 160 km, find the cluster size of the cellular system.

Solution:  $R = 10$  km

$D = 160$  km

$N^2 = 16 \times 16/3$

Hence, the cluster size is  $N = 16/\sqrt{3}$